

Transcriptomic Pathway Activity Identifies Molecularly Distinct Subgroups of IDH Wild- Type Glioblastoma with Different Mutation Landscapes and Clinical Outcomes

Hanna Dmochowska

Date: 29.05.2026

Background

Glioblastoma multiforme (GBM; WHO Grade IV astrocytoma) is the most lethal primary brain tumour in adults, with median overall survival of approximately 14.6 months despite maximal safe resection, radiotherapy, and temozolomide (the Stupp protocol) (Stupp et al., 2005; Tan et al., 2020). These outcomes reflect profound molecular heterogeneity that limits the efficacy of uniform treatment, motivating molecular stratification strategies. Integrating RNA sequencing (RNA-seq) with somatic mutation data is a principled route to such stratification: each data layer has complementary limitations (individual mutations occur in only a minority of patients; raw expression creates a severe $p \gg n$ dimensionality problem for survival modelling with $n \sim 100$), and combining them connects upstream genomic events to their downstream transcriptomic consequences (Tambi et al., 2024). Compressing $\sim 20,000$ -gene expression to 14 PROGENy pathway activity scores (Schubert et al., 2018) addresses this: scores represent biologically interpretable causal pathway activation states and have been shown to yield more stable prognostic models than gene-level features across 33 TCGA cancer types (Zheng et al., 2021). Analyses combine unsupervised methods (Kaplan-Meier survival estimation, k-means clustering, correlation profiling) with supervised methods (Cox proportional hazards regression, elastic net regularised Cox regression) to identify patterns without and with using survival as a training signal respectively.

IDH mutation is the strongest prognostic factor in diffuse gliomas (Brat et al., 2015) and must be treated as a prognostic baseline; IDH-mutant and wild-type GBMs are now classified as distinct disease entities (WHO 2021). Within IDH wild-type GBM ($\sim 90\%$ of primary cases), the TCGA previously identified three transcriptomic subtypes - Classical (EGFR/MAPK-driven), Mesenchymal (*NF1*/inflammatory), and Proneural (*PDGFRA*-altered) - with distinct mutation landscapes and clinical outcomes (Verhaak et al., 2010; Brennan et al., 2013). This motivates the research question: Do transcriptomic pathway activity patterns identify molecularly distinct subgroups of IDH wild-type GBM with different somatic mutation landscapes and clinical outcomes? RNA-seq (RSEM z-scores), somatic mutation, and clinical data from the TCGA GBM PanCan Atlas 2018 cohort (cBioPortal; $n = 120$ aligned patients) were used. After IDH-baseline analysis, the cohort was restricted to IDH wild-type patients ($n = 109$). k-means clustering on PROGENy scores identified transcriptomic subtypes, which were characterised by mutation enrichment profiles (Fisher exact tests) and survival outcomes (Kaplan-Meier, Cox regression, elastic net regularised Cox).

Results

1. Dataset Validation and IDH Prognostic Baseline

After aligning clinical, RNA-seq, and somatic mutation data, 120 patients were retained. The Kaplan-Meier (KM) survival curve for the full cohort (Figure 1, left panel; unsupervised) shows a median OS of approximately 12-14 months, consistent with the Stupp protocol benchmark (Stupp et al., 2005). IDH mutation rate was approximately 8% (10/120), consistent with primary GBM cohorts (Brat et al., 2015). IDH-mutant patients showed substantially longer survival (log-rank $p = 0.026$; Figure 1, right panel), replicating a benchmark result and confirming correct biological signal in the dataset. All subsequent analyses were restricted to the 109 IDH wild-type patients to

ensure biological homogeneity and avoid confounding by the fundamentally different transcriptional landscape of IDH-mutant GBM.

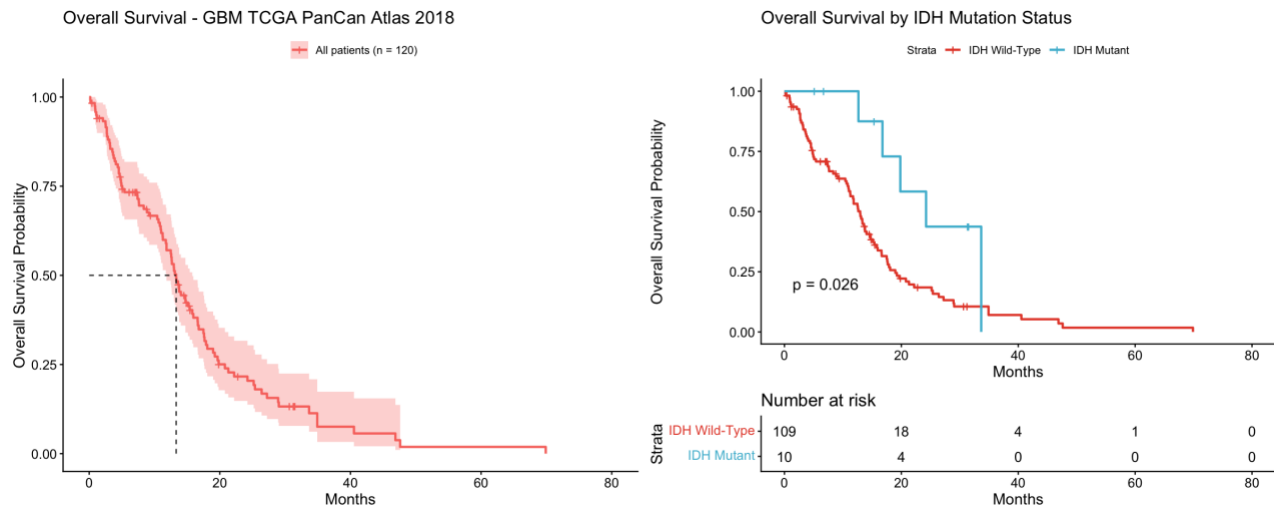


Figure 1. Kaplan-Meier survival curves for the TCGA GBM cohort. Left panel (A): Overall survival (n = 120); median OS approximately 12-14 months. Right panel (B): Survival stratified by IDH mutation status; red = IDH wild-type (n = 110), blue = IDH mutant (n = 10), log-rank p = 0.026.

2. Somatic Mutation Landscape of the GBM Cohort

To characterise the overall somatic mutation landscape, maftools was used to visualise mutation frequencies across the full TCGA GBM mutation dataset. The oncoplot (Figure 2) shows the top mutated genes: *PTEN* (33%), *TP53* (31%), *TTN* (26%), *EGFR* (23%), *NFI* (12%), and *RBI* (10%), consistent with the canonical GBM driver gene landscape reported by the TCGA (Brennan et al., 2013; TCGA, 2008). Missense mutations predominate, with copy number alterations also prominent for *PTEN* and *EGFR*. Somatic interaction analysis (Figure 3) identified significant co-occurrence of *PTEN-RBI* and *TTN-PTEN* mutations ($p < 0.05$), as well as borderline mutual exclusivity between *EGFR* and *TP53* ($p < 0.1$), consistent with these genes acting in parallel oncogenic pathways where co-mutation provides no additional selective advantage (Brennan et al., 2013).

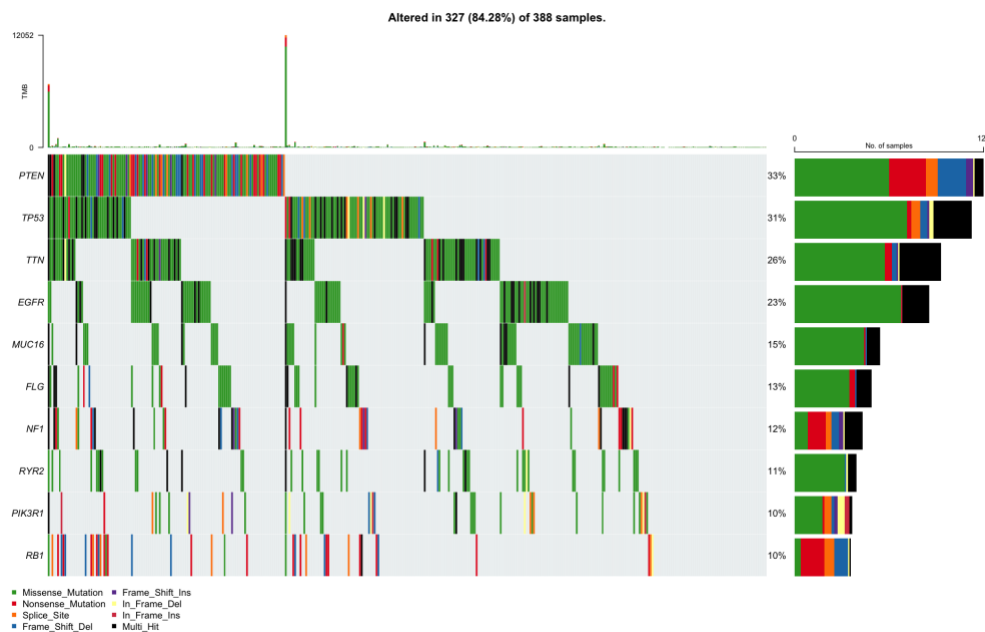


Figure 2. Oncoplot of the top 10 somatically mutated genes in the TCGA GBM cohort. Generated using maftools (GBM_survival_analysis.R Section 3b). Each column represents a patient; each row a gene. Colour indicates mutation type (missense, nonsense, frame shift, splice site, multi-hit). Top mutated genes: PTEN (33%), TP53 (31%), TTN (26%), EGFR (23%), NF1 (12%), RB1 (10%). Mutation frequency shown on the right. TMB shown above each column.

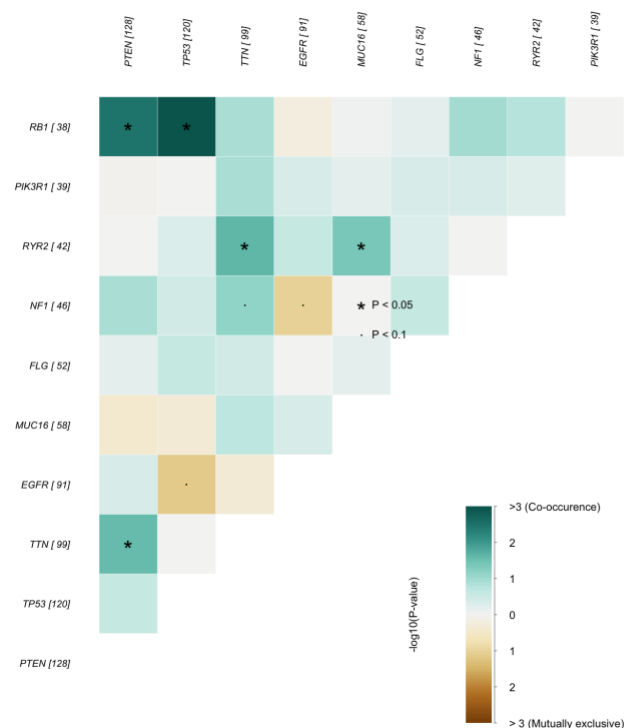


Figure 3. Somatic interaction analysis for the top 10 mutated genes. Co-occurrence (blue) and mutual exclusivity (red) between gene pairs estimated by Fisher exact test. Asterisk (*): $p < 0.05$; full stop (.): $p < 0.1$. Significant co-occurrence: PTEN-RB1 (*) and TTN-PTEN (*). Borderline mutual exclusivity: EGFR-TP53 (.), consistent with parallel oncogenic pathway activity.

3. PROGENy Pathway Co-activation Structure

Pathway activities were inferred from RNA-seq of the 109 IDH wild-type patients using PROGENy, yielding 14 continuous activity scores per patient (Schubert et al., 2018). Pairwise Pearson correlations were visualised as a hierarchically clustered correlogram (Figure 4; unsupervised). Two major co-activation clusters emerged: an RTK/mitogenic cluster (EGFR, MAPK) and an inflammatory/mesenchymal cluster (JAK-STAT, Trail, p53, TNFa, NFkB). The strong negative correlation between these clusters indicates that RTK-driven and inflammatory signalling states are mutually exclusive across patients, consistent with the existence of biologically distinct transcriptomic subtypes (Verhaak et al., 2010).

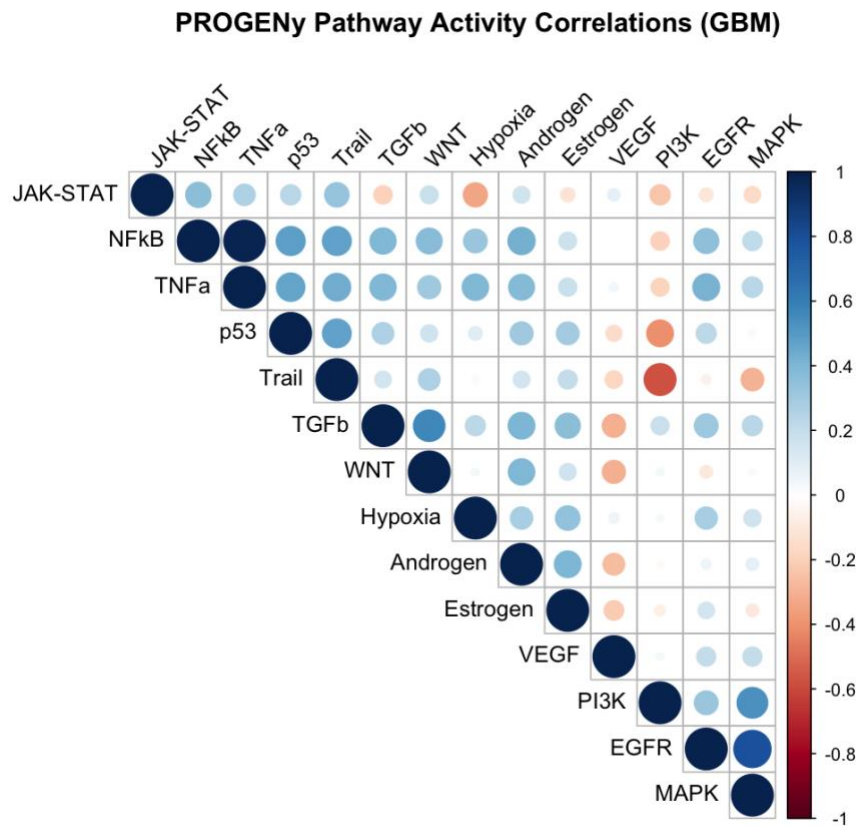


Figure 4. PROGENy pathway activity correlation matrix (IDH wild-type GBM, n = 109). Hierarchically clustered Pearson correlations. RTK/mitogenic cluster (EGFR, MAPK) and inflammatory/mesenchymal cluster (JAK-STAT, Trail, p53, TNFa, NFkB) are negatively correlated.

4. Unsupervised Subtype Discovery by Pathway Clustering

K-means clustering (*stats::kmeans*; *nstart* = 50; *iter.max* = 200; *set.seed*(42)) was applied to the 14 PROGENy scores. The elbow plot (Figure 5) was consistent with *k* = 3, selected in combination with prior TCGA subtyping literature (Verhaak et al., 2010; Brennan et al., 2013). Three subtypes were identified (Figure 6): Subtype 1 (*n* = 22), globally hyperactivated with the highest EGFR pathway activity (mean = 1.04); Subtype 2 (*n* = 37), Trail/JAK-STAT-low with moderate RTK signalling; Subtype 3 (*n* = 50), PI3K/VEGF-elevated. These profiles closely match the Classical,

Mesenchymal, and Proneural/Neural subtypes of Verhaak et al. (2010), independently validating the PROGENy dimensionality reduction.

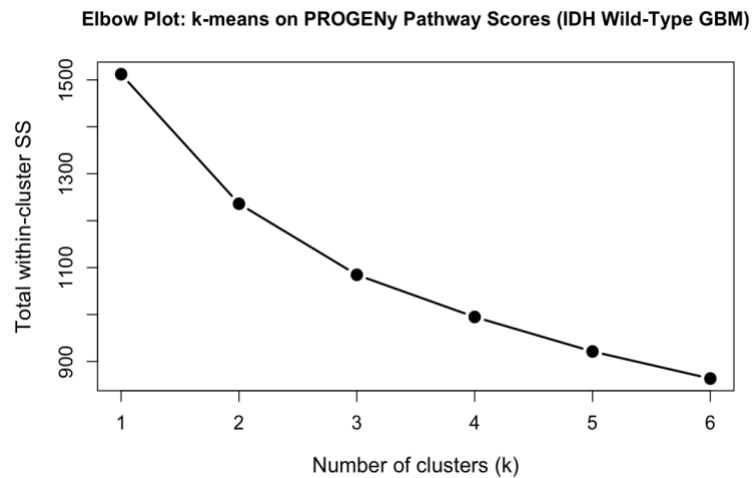


Figure 5. Elbow plot for k-means clustering on PROGENy pathway scores (IDH wild-type GBM). Total within-cluster sum of squares plotted for $k = 1$ to 6. The elbow at $k = 3$ supports selection of three subtypes, consistent with prior TCGA subtyping literature.

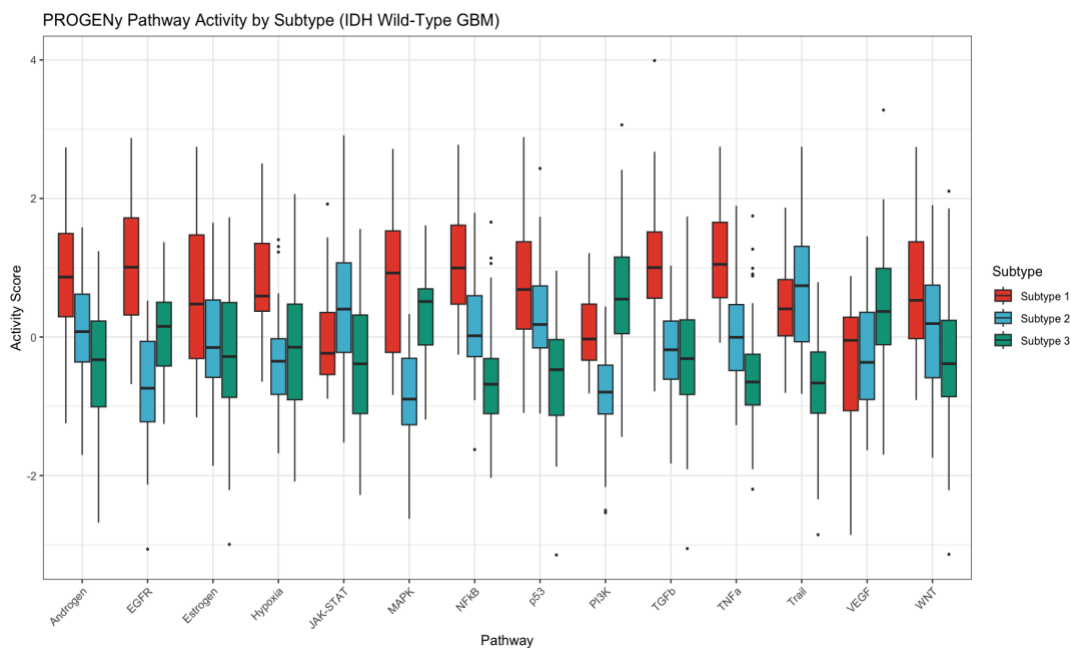


Figure 6. PROGENy pathway activity distributions by transcriptomic subtype. Boxplots of all 14 pathway activity scores for each of the three k-means subtypes ($n = 109$). Subtype 1 ($n = 22$): globally hyperactivated, highest EGFR pathway activity (mean = 1.04); Subtype 2 ($n = 37$): Trail/JAK-STAT-low, moderate RTK; Subtype 3 ($n = 50$): PI3K/VEGF-elevated.

5. Somatic Mutation Landscapes of Transcriptomic Subtypes

Driver gene mutation frequencies were computed per subtype and tested by two-sided Fisher exact test (Figure 7). The key finding was a paradoxical discordance in Subtype 1: despite the highest EGFR pathway activity (mean = 1.04), it showed the lowest *EGFR* somatic mutation frequency (9.1%), significantly depleted relative to the rest of the cohort (Fisher $p = 0.0011$). EGFR pathway hyperactivation in Subtype 1 is instead driven by *PTEN* loss (54.5%), which disinhibits the PI3K/AKT axis, and *NF1* enrichment (31.8%, Fisher $p = 0.018$), which derepresses RAS/MAPK activity - both activating EGFR downstream signalling without a direct *EGFR* mutation. By contrast, Subtype 2 showed the highest *EGFR* somatic mutation rate (48.6%) yet lower pathway activity, suggesting these mutations are not fully functionally activating or are offset by downstream regulatory mechanisms. This genomic-transcriptomic discordance is the central finding of this study: the two data layers encode non-redundant, sometimes counter-intuitive biological information, demonstrating that integration is necessary for accurate characterisation of GBM subtype biology (Brennan et al., 2013; Verhaak et al., 2010).

Driver Mutation Frequency by Transcriptomic Subtype (IDH Wild-Type GBM)

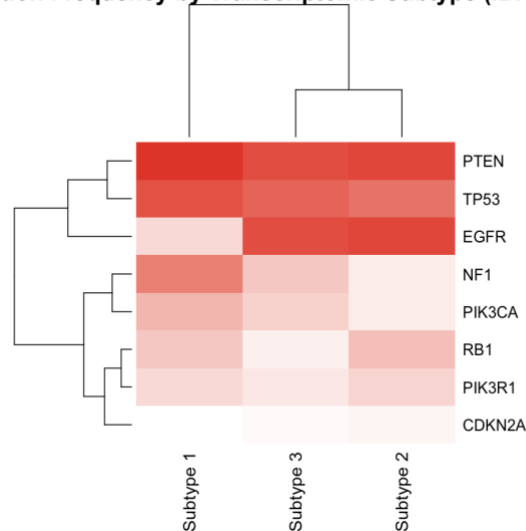


Figure 7. Driver gene mutation frequency heatmap by transcriptomic subtype. Rows: driver genes; columns: subtypes. Colour intensity: mutation frequency (0-1 scale). Key finding: Subtype 1 (EGFR pathway-high) shows paradoxically low EGFR mutation frequency (9.1%, Fisher $p = 0.0011$, depleted), driven instead by PTEN loss (54.5%) and NF1 enrichment (31.8%, Fisher $p = 0.018$). Subtype 2 shows highest EGFR mutation rate (48.6%) despite lower pathway activity.

The cross-omics correlation matrix (Figure 8) provides direct visual confirmation of this discordance: *EGFR_mut* shows near-zero correlation with EGFR pathway score (pale, small circle), demonstrating that having an *EGFR* somatic mutation does not predict EGFR pathway activity at the transcriptomic level. This independently validates the Fisher exact test result and illustrates why mutation and pathway data are non-redundant layers requiring integration (Brennan et al., 2013; Tambi et al., 2024).

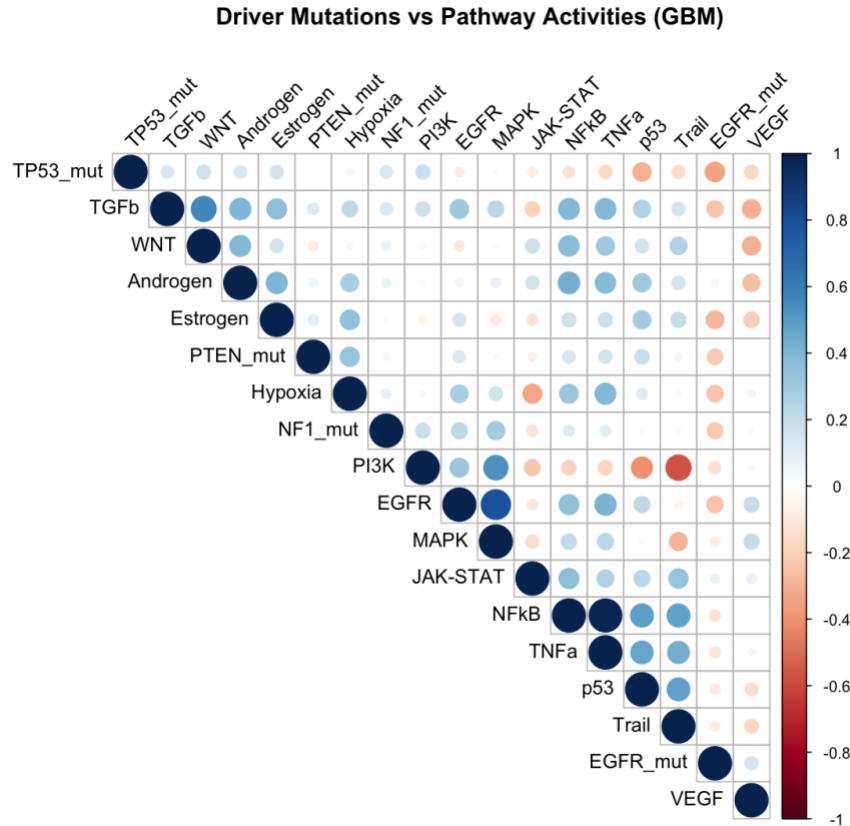


Figure 8. Cross-omics Pearson correlation matrix: somatic mutation indicators vs. PROGENy pathway scores. Rows: binary mutation indicators for 8 driver genes; columns: 14 PROGENy pathway scores. Circle size and colour intensity encode correlation magnitude. Key observation: EGFR_mut has near-zero correlation with EGFR pathway score, directly confirming genomic-transcriptomic discordance.

6. Subtype Survival Analysis

KM survival curves by transcriptomic subtype (Figure 9; unsupervised) did not reach significance (log-rank $p = 0.18$), consistent with limited statistical power in $n = 109$ and uniformly poor prognosis across IDH wild-type GBM. This does not contradict the molecular distinctiveness demonstrated in Section 5: the subtypes differ significantly in mutation landscape (Fisher $p = 0.0011$ and 0.018) and represent biologically real entities. The survival signal is recovered by more sensitive targeted analyses in Section 7.

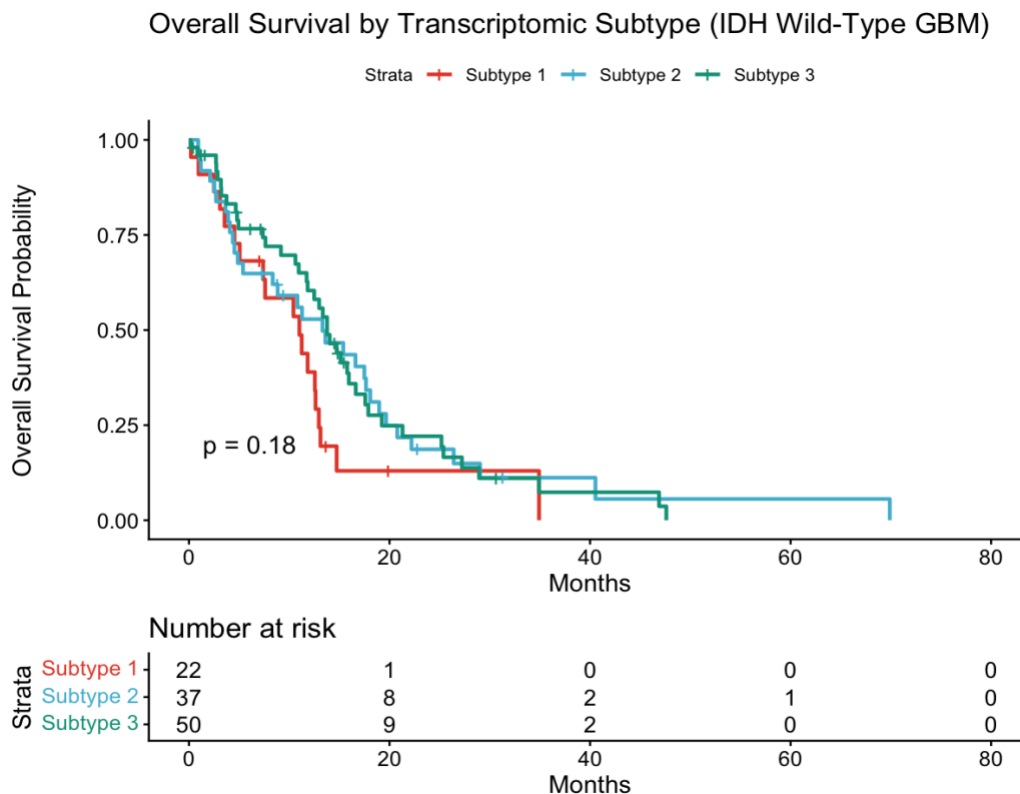


Figure 9. Kaplan-Meier survival curves by transcriptomic subtype (IDH wild-type GBM, n = 109). Three subtypes derived from k-means clustering on PROGENy scores. Log-rank $p = 0.18$ (not statistically significant). Risk table: numbers at risk at 20-month intervals.

7. EGFR Pathway Extremes and Supervised Elastic Net Validation

A scan of all interior EGFR pathway score thresholds (unsupervised) identified a subgroup of 8 patients with extreme hyperactivation (score > 1.37) with near-uniform mortality within 10 months (log-rank $p = 0.0011$; Figure 14, left panel). As this threshold was data-adaptively optimised, the p-value is exploratory and subject to multiple-testing inflation; independent validation is required. Critically, consistent with Section 5, this extreme-EGFR subgroup is not defined by *EGFR* somatic mutation but by *PTEN* loss and *NF1* mutations - a finding that mutation profiling alone would miss.

As a prerequisite to integration, individual Cox proportional hazards models were fitted for gene expression and mutation features separately. All individual gene expression predictors were non-significant (global $p = 0.165$; Figure 10) and all individual mutation predictors were likewise non-significant (global $p = 0.393$; Figure 11), motivating the move to regularised multi-feature models. As supervised validation, elastic net regularised Cox regression was applied to three feature sets (mutations only, pathways only, integrated), with alpha scanned across 11 values from 0 to 1 and lambda tuned by 10-fold cross-validated C-index (set.seed(42)). The optimal alpha was 0 (pure ridge), retaining all 22 integrated features. In the final elastic net integrated model, *EGFR* pathway score emerged as the strongest positive prognostic feature (HR = 2.01, 95% CI 1.12-3.61, $p = 0.019$), while PI3K pathway showed a protective association (HR = 0.62, 95% CI 0.41-0.94, $p = 0.024$; Figure 12). This EGFR-PI3K opposition mirrors the negative correlation seen in the

pathway co-activation structure (Figure 4) and the genomic discordance observed at the mutation level (Section 5). In the nested Cox LRT multivariate model, *RBI* mutation was the only individually significant predictor ($p = 0.042$), reinforcing the value of combining mutation and pathway features. A nested Cox LRT confirmed that adding pathway scores to a mutations-only model significantly improved fit ($\text{chi-sq} = 11.25$, $\text{df} = 5$, $p = 0.047$). Cross-validated C-indices were: mutations = 0.554, pathways = 0.554, integrated = 0.580 (Figure 13; Table 1), showing modest but consistent improvement from integration. The integrated risk score stratified survival at the median (log-rank $p = 0.00024$; Figure 14, right panel).

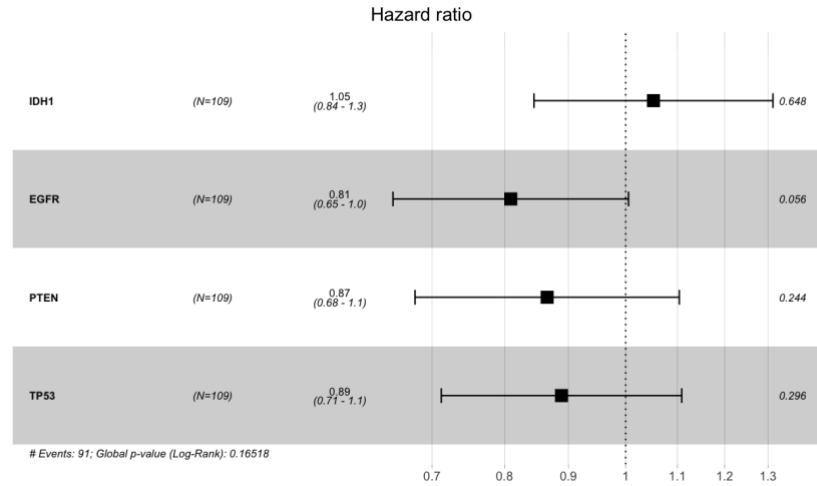


Figure 10. Individual gene expression Cox model: hazard ratios (IDH wild-type GBM, $n = 109$). Forest plot for individual univariate Cox regression on gene expression z-scores. All features non-significant; global likelihood ratio test $p = 0.165$. This motivates multi-feature regularised modelling.

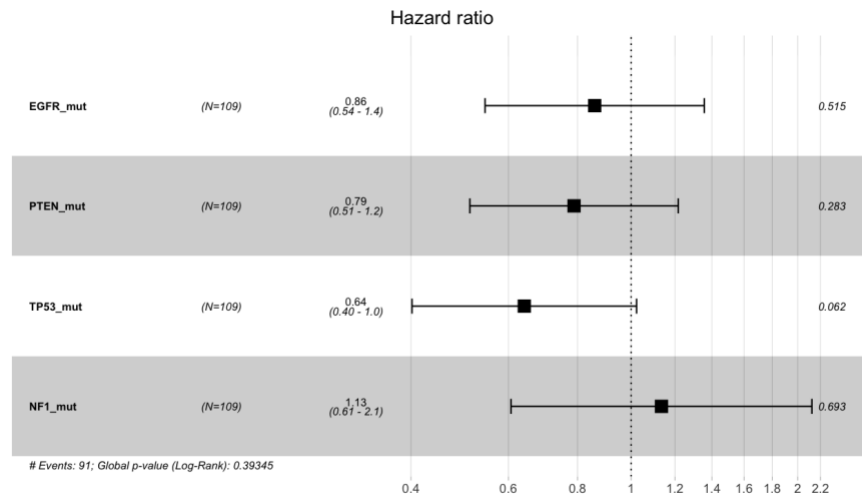


Figure 11. Individual mutation Cox model: hazard ratios (IDH wild-type GBM, $n = 109$). Forest plot for individual univariate Cox regression on binary mutation indicators. All features non-significant; global likelihood ratio test $p = 0.393$. Confirms that no single mutation alone is sufficient to stratify GBM survival, motivating integration. Generated by GBM_survival_analysis.R Section 8.

Feature	Type	Ridge coefficient
EGFR pathway	Pathway score	+0.259
NF1_mut	Mutation	+0.215
CDKN2A_mut	Mutation	+0.122
TGFb pathway	Pathway score	+0.099
Androgen pathway	Pathway score	+0.081
PIK3CA_mut	Mutation	+0.080
Hypoxia pathway	Pathway score	+0.079
Trail pathway	Pathway score	+0.078
Estrogen pathway	Pathway score	+0.062
PIK3R1_mut	Mutation	+0.015
NFkB pathway	Pathway score	+0.010
WNT pathway	Pathway score	-0.039
TNFa pathway	Pathway score	-0.045
JAK-STAT pathway	Pathway score	-0.045
MAPK pathway	Pathway score	-0.051
EGFR_mut	Mutation	-0.031
VEGF pathway	Pathway score	-0.103
p53 pathway	Pathway score	-0.092
PTEN_mut	Mutation	-0.188
PI3K pathway	Pathway score	-0.235
TP53_mut	Mutation	-0.317
RB1_mut	Mutation	-0.674

Table 2. Elastic net ridge regression coefficients for all 22 integrated features. Sorted by coefficient magnitude (descending positive to most negative). Positive values: associated with increased hazard; negative values: associated with reduced hazard. Features are shrunk but not zeroed by ridge ($\alpha = 0$); coefficient magnitude reflects relative contribution. Key features: EGFR pathway (+0.259, highest positive), RB1_mut (-0.674, largest magnitude overall), TP53_mut (-0.317), PI3K pathway (-0.235). Note EGFR_mut coefficient (-0.031) is near-zero, consistent with the genomic-transcriptomic discordance finding (Figure 8).

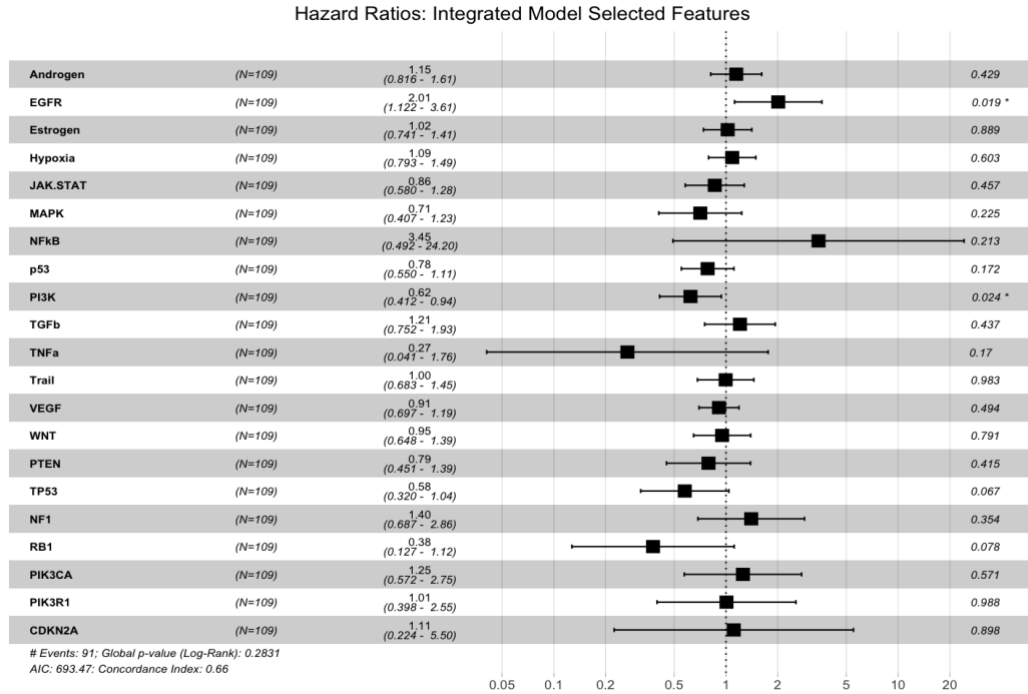


Figure 12. Elastic net integrated Cox model: hazard ratios for retained features. Forest plot for the pure ridge ($\alpha = 0$) elastic net Cox regression on 22 integrated features (14 PROGENy pathway scores + 8 mutation indicators) in the IDH wild-type GBM cohort ($n = 109$). EGFR pathway: HR = 2.01 (95% CI 1.12-3.61), $p = 0.019^*$. PI3K pathway: HR = 0.62 (95% CI 0.41-0.94), $p = 0.024^*$. All 22 features retained; concordance index = 0.66.

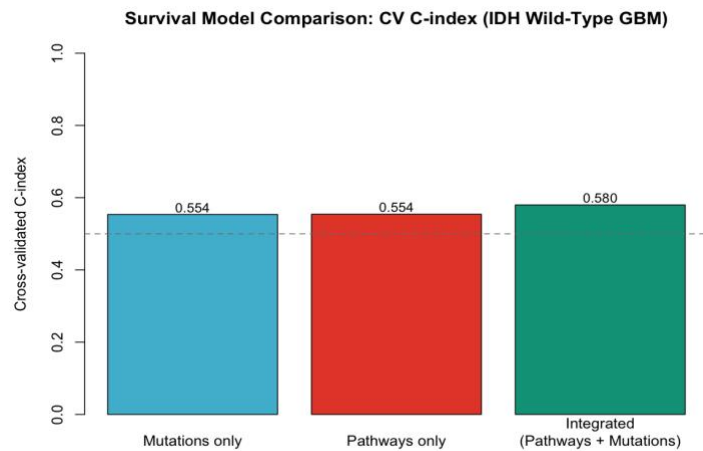
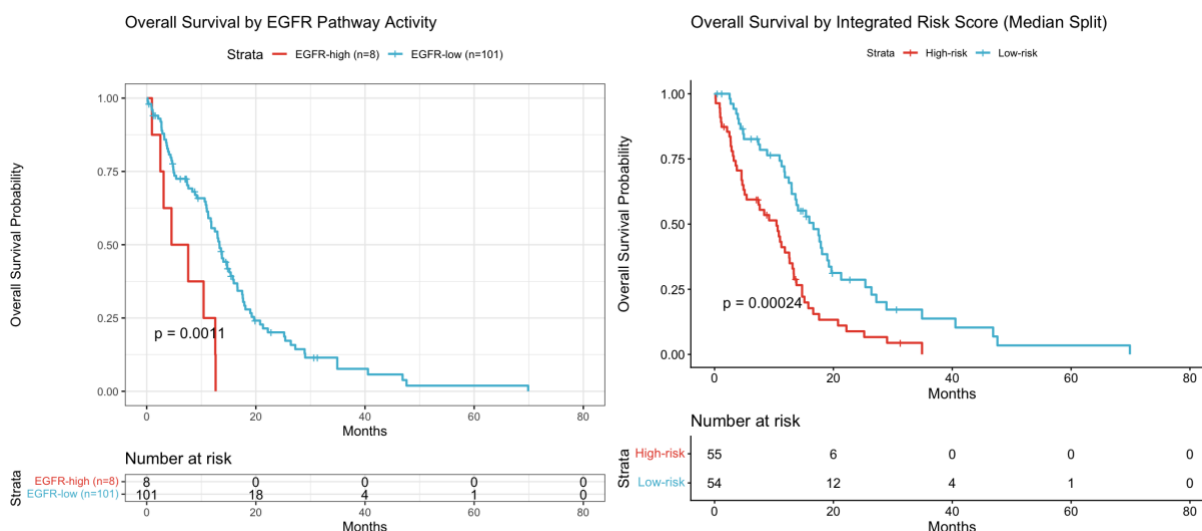


Figure 13. Cross-validated concordance indices for three elastic net Cox models. 10-fold CV C-indices: mutations only = 0.554, pathways only = 0.554, integrated = 0.580. Dashed line at $C = 0.5$ (no discrimination). The integrated model consistently outperforms single-omic models.

Model	Features	CV C-index	Note
Mutations only (elastic net)	Binary driver mutation indicators	0.554	10-fold CV; alpha = 0.5
Pathways only (elastic net)	14 PROGENy pathway scores	0.554	10-fold CV; alpha = 0.5
Integrated (elastic net)	14 PROGENy scores + mutation indicators	0.580	10-fold CV; alpha = 0 (pure ridge)

Table 1. Cross-validated concordance index for three elastic net Cox models (IDH wild-type GBM). CV C-index: 10-fold cross-validated; unbiased estimate of generalisation. C = 0.5: no discrimination; C = 1.0: perfect. Mutations-only and pathways-only: alpha = 0.5; integrated model: alpha = 0 (pure ridge, selected by alpha scan). IDH wild-type cohort; n = 109. Nested Cox LRT confirms pathway scores add significant fit beyond mutations alone (chi-sq = 11.25, df = 5, p = 0.047).



Conclusion

Transcriptomic pathway activity patterns do identify molecularly and clinically distinct subgroups of IDH wild-type GBM that cannot be resolved by somatic mutation data alone. The central finding is a paradoxical genomic-transcriptomic discordance: Subtype 1, with the highest EGFR pathway activity (mean = 1.04), shows significant depletion of *EGFR* somatic mutations (9.1%, p = 0.0011) and is instead driven by *PTEN* loss (54.5%) and *NF1* enrichment (31.8%, p = 0.018) - a mechanism invisible to mutation profiling alone. In the supervised elastic net model, *EGFR* pathway score is the dominant prognostic feature (HR = 2.01, p = 0.019) with PI3K showing a protective opposing effect (HR = 0.62, p = 0.024), directly reflecting the EGFR-PI3K anticorrelation structure observed

unsupervised (Figure 4). The formal LRT confirms pathway scores add significant prognostic information beyond mutation status ($p = 0.047$), and the integrated risk score identifies patients with markedly different outcomes ($p = 0.00024$), providing direct supervised validation of the multi-omics integration approach. These results align with emerging evidence that multi-omics integration consistently outperforms single-layer models in cancer survival prediction (Fan et al., 2025; Zheng et al., 2021).

Several limitations apply. The IDH wild-type cohort ($n = 109$) limits statistical power for subtype-gene associations; Fisher exact tests may be underpowered for low-frequency alterations. The maftools oncoplot uses the full TCGA mutation file ($n = 388$) to provide a broad landscape overview, while subtype-level analyses use only the 109-patient aligned cohort. The EGFR threshold p-value is subject to multiple-testing inflation from data-adaptive selection and requires validation in an independent cohort (e.g. CGGA or GLASS). Bulk RNA-seq does not resolve intra-tumour or microenvironmental heterogeneity, which are increasingly recognised as important in GBM (Tambi et al., 2024; Lucas et al., 2025). K-means imposes hard cluster boundaries on a continuous signalling landscape; probabilistic approaches may better capture subtype uncertainty. All three elastic net models showed only modest C-index values (0.554-0.580), reflecting the inherent prognostic difficulty in this homogeneously poor-prognosis cohort.

Code Availability

All code is provided in GBM_survival_analysis.R. The script is structured into clearly labelled sections corresponding to the Results subsections above, with inline comments justifying all methodological choices and hyperparameter settings. Data are publicly available via cBioPortal (TCGA GBM PanCan Atlas 2018; <https://www.cbioportal.org>).

References

- Brat, D. J., Verhaak, R. G. W., Aldape, K. D., Yung, W. K. A., Salama, S. R., Cooper, L. A. D., ... Miller, R. (2015). Comprehensive, Integrative Genomic Analysis of Diffuse Lower-Grade Gliomas. *New England Journal of Medicine*, 372(26), 2481–2498.
<https://doi.org/10.1056/nejmoa1402121>
- Brennan, Cameron W., Verhaak, Roel G. W., McKenna, A., Campos, B., Nounshmehr, H., Salama, Sofie R., ... Ciriello, G. (2013). The Somatic Genomic Landscape of Glioblastoma. *Cell*, 155(2), 462–477. <https://doi.org/10.1016/j.cell.2013.09.034>
- Fan, X., He, Z., Guo, J., Bu, D., Han, D., Qu, X., ... Guo, J. (2025). Leveraging TME features and multi-omics data with an advanced deep learning framework for improved Cancer survival prediction. *Scientific Reports*, 15(1). <https://doi.org/10.1038/s41598-025-98565-0>
- Lucas, C.-H. G., Al-Adli, N. N., Young, J. S., Gupta, R., & Morshed, R. A. (2025, January 1). Longitudinal multimodal profiling of IDH-wildtype glioblastoma reveals the molecular evolution and cellular phenotypes underlying prognostically different treatment responses. Retrieved from <https://doi.org/10.1093/neuonc/noae214>
- Schubert, M., Klinger, B., Klünemann, M., Sieber, A., Uhlitz, F., Sauer, S., ... Saez-Rodriguez, J. (2018). Perturbation-response genes reveal signaling footprints in cancer gene expression. *Nature Communications*, 9(1). <https://doi.org/10.1038/s41467-017-02391-6>
- Stupp, R., Mason, W. P., van den Bent, M. J., Weller, M., Fisher, B., Taphoorn, M. J. B., ... Mirimanoff, R. O. (2005). Radiotherapy plus Concomitant and Adjuvant Temozolomide for Glioblastoma. *New England Journal of Medicine*, 352(10), 987–996.
<https://doi.org/10.1056/nejmoa043330>
- Tambi, R., Zehra, B., Vijayakumar, A., Satsangi, D., Uddin, M., & Berdiev, B. K. (2024). Artificial Intelligence and Omics in Malignant Gliomas. *Physiological Genomics*, 56(12), 876–895. <https://doi.org/10.1152/physiolgenomics.00011.2024>

- Tan, A. C., Ashley, D. M., López, G. Y., Malinzak, M., Friedman, H. S., & Khasraw, M. (2020). Management of glioblastoma: State of the art and future directions. *CA: A Cancer Journal for Clinicians*, 70(4), 299–312. <https://doi.org/10.3322/caac.21613>
- The Cancer Genome Atlas Research Network. (2008). Comprehensive genomic characterization defines human glioblastoma genes and core pathways. *Nature*, 455(7216), 1061–1068. <https://doi.org/10.1038/nature07385>
- Verhaak, R. G. W., Hoadley, K. A., Purdom, E., Wang, V., Qi, Y., Wilkerson, M. D., ... Feiler, H. S. (2010). Integrated Genomic Analysis Identifies Clinically Relevant Subtypes of Glioblastoma Characterized by Abnormalities in PDGFRA, IDH1, EGFR, and NF1. *Cancer Cell*, 17(1), 98–110. <https://doi.org/10.1016/j.ccr.2009.12.020>
- Zheng, X., Amos, C. I., & Frost, H. R. (2021). Pan-cancer evaluation of gene expression and somatic alteration data for cancer prognosis prediction. *BMC Cancer*, 21(1). <https://doi.org/10.1186/s12885-021-08796-3>